

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1689

Charles D. Lamb Energy Center 138 kV
February 25, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1689. The Interconnection Customer proposes a 104.1 MW battery energy storage facility interconnecting at the Charles D. Lamb Energy Center 138 kV substation in the Pennsylvania area. The request was received on March 07, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 610 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 16 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$40,432,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: TALENENERGY MARTINS CRK LLC LOCK HAVEN to EL CERRITO STA G 138kV
Contingency: FT HUMBUG to CORONET 500kV
Loading: 138.2% (Rating: 235 MVA)
DFAX: 14.2%
Pre-project loading: 132.0%
2. Facility: TALENENERGY MARTINS CRK LLC LOCK HAVEN to TAYLOR FLATS 138kV Line
Contingency: CORONET to FT HUMBUG 500kV
Loading: 122.5% (Rating: 265 MVA)
DFAX: 14.2%
Pre-project loading: 117.0%
3. Facility: TAYLOR FLATS to PASTILLO 5801 138kV
Contingency: CORNET to FT HUMBUG 500kV
Loading: 121.1% (Rating: 202 MVA)
DFAX: 8.0%
Pre-project loading: 117.0%
4. Facility: GEORGE STREET SUBSTATION to CALDEN 138kV
Contingency: CHARITY CHURCH to GEORGE ST 138kV
Loading: 117.1% (Rating: 194 MVA)
DFAX: 5.4%
Pre-project loading: 114.2%
5. Facility: EL CERRITO STA G 138/1 kV Transformer
Contingency: SCOTT to SHAWVILLE STATION 345kV
Loading: 116.5% (Rating: 325 MVA)
DFAX: 36.2%
Pre-project loading: 104.9%

6. Facility: PASTILLO 5801 345-1 kV Auto
Contingency: MONTOUR STATION to TALENENERGY MARTINS CRK LLC WILLIAMSPORT 345kV Line
Loading: 113.1% (Rating: 369 MVA)
DFAX: 5.7%
Pre-project loading: 111.5%
7. Facility: KNIGHTS LANDING to VOLCNO 138kV
Contingency: FORT HUMBUG to CORONET 500kV
Loading: 111.1% (Rating: 182 MVA)
DFAX: 39.4%
Pre-project loading: 88.6%
8. Facility: GEORGE ST to CHARITY CHURCH 138kV
Contingency: CHARITY CHURCH to CALDEN 138kV
Loading: 110.8% (Rating: 213 MVA)
DFAX: 5.6%
Pre-project loading: 108.1%
9. Facility: PASTILLO 5801 138-1 kV Transformer
Contingency: MONTOUR STATION SUBSTATION to TALENENERGY MARTINS CRK LLC WILLIAMSPORT 345kV
Loading: 110.4% (Rating: 378 MVA)
DFAX: 5.7%
Pre-project loading: 108.8%
10. Facility: CALDEN to CHARITY CHURCH 138kV
Contingency: CHARITY CHURCH to GEORGE ST 138kV
Loading: 109.0% (Rating: 211 MVA)
DFAX: 5.4%
Pre-project loading: 106.3%
11. Facility: NORTHSIDE DRIVE to VOLCANO 138kV
Contingency: FT HUMBUG to CORONET 500kV
Loading: 105.7% (Rating: 246 MVA)
DFAX: 8.5%
Pre-project loading: 102.1%
12. Facility: BUXTON to NORTHSIDE DRIVE 138kV
Contingency: CORONET to FORT HUMBUG 500kV
Loading: 105.4% (Rating: 250 MVA)
DFAX: 8.5%
Pre-project loading: 101.8%
13. Facility: BUXTON to YEAGERTOWN 138kV
Contingency: CORONET to FT HUMUG SUBSTATION 500kV
Loading: 104.8% (Rating: 258 MVA)
DFAX: 8.5%
Pre-project loading: 101.4%
14. Facility: VOLCANO to WALNUT CREEK 138kV
Contingency: FT HUMBUG to CORONET 500kV
Loading: 103.7% (Rating: 738 MVA)
DFAX: 25.2%
Pre-project loading: 100.2%
15. Facility: GE ALACHUA to CHARLES D. LAMB ENERGY CTR 138kV
Contingency: CORONET to FT HUMBUG 500kV Line
Loading: 101.8% (Rating: 230 MVA)
DFAX: 60.6%
Pre-project loading: 74.5%

16. Facility: EL CERRITO STA G 345/1kV Transformer
 Contingency: SHAWVILLE STATION SUBSTATION to SCOTT 345kV L
 Loading: 100.4% (Rating: 384 MVA)
 DFAX: 36.2%
 Pre-project loading: 90.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
TALENENERGY MARTINS CREEK LLC LOCK HAVEN to EL CERRITO STA G 138kV line	Rebuild line with higher-capacity conductor	\$8,540,000	21% (\$1,807,000)
TAYLOR FLATS to TALENENERGY MARTINS CRK LLC LOCK HAVEN 138kV	Rebuild line with higher-capacity conductor	\$4,172,000	21% (\$883,000)
TAYLOR FLATS to PASTILLO 5801 138kV	Rebuild line with higher-capacity conductor	\$29,411,000	12% (\$3,413,000)
GEORGE STREET to CALDEN 138kV	Rebuild line with higher-capacity conductor	\$1,320,000	33% (\$439,000)
EL CERRITO STA G 138 kV/1 kV XFMR	Replace transformer with higher-rated unit	\$4,844,000	35% (\$1,681,000)
PASTILLO 5801 345-1 kV Auto	Replace transformer with higher-rated unit	\$19,946,000	5% (\$990,000)
KNIGHTS LANDING to VOLCANO 138kV	Rebuild line with higher-capacity conductor	\$12,436,000	100% (\$12,436,000)
CHARITY CHURCH to GEORGE STREET 138kV	Rebuild line with higher-capacity conductor	\$9,697,000	33% (\$3,229,000)
PASTILLO 5801 138/1 kV TX	Replace transformer with higher-rated unit	\$3,541,000	5% (\$176,000)
CALDEN to CHARITY CHURCH 138kV	Reconductor line	\$1,559,000	33% (\$519,000)
NORTHSIDE DRIVE to VOLCANO 138kV	Reconductor line	\$1,718,000	23% (\$391,000)
NORTHSIDE DRIVE to BUXTON 138kV	Reconductor line	\$1,708,000	23% (\$388,000)
BUXTON to YEAGERTOWN SUBSTATION 138kV	Reconductor line	\$3,123,000	23% (\$710,000)
VOLCANO to WALNUT CREEK 138kV	Reconductor line	\$2,130,000	44% (\$935,000)
GE ALACHUA to CHARLES D. LAMB ENERGY CTR 138kV Line	Reconductor line	\$8,400,000	100% (\$8,400,000)
EL CERRITO STA G 345-1 kV Auto	Replace transformer with higher-rated unit	\$11,624,000	35% (\$4,035,000)
	Total allocated to project		\$40,432,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.